TaPTaP Game Engine Hackathon — Checkpoint 1

Engine Blueprint

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This document defines the core architecture of a reusable web based learning game engine designed for the TaPTaP Game Engine Hackathon. The engine is JSON driven and modular, allowing multiple learning games to be generated using configuration files rather than hardcoded game logic.

1. Core Engine Loop

CORE ENGINE LOOP

Start Engine
↓
Load JSON Game Configuration
↓
Initialize Game Components
↓
Start Game Loop WHILE Game Running:
→ Capture Player Input
→ Update Game State
→ Apply Game Rules
→ Update Score / Progress
→ Render Game Frame
→ Check Win / Loss Condition END LOOP
Display Game Result

2. JSON Configuration Structure

```
JSON CONFIGURATION STRUCTURE (game-config.json)

{
  "game": {
    "title": "Math Tap Challenge",
    "version": "1.0",
    "maxLevels": 3
  },
  "player": {
    "lives": 3,
    "startingScore": 0
  },

  "timer": {
    "enabled": true,
    "timeLimitSeconds": 60
  },

  "scoring": {
    "correctAnswer": 10,
    "wrongAnswer": -5,
    "levelCompletionBonus": 50
  },
```

```
  "levels": [
    { "levelId": 1, "difficulty": "easy", "questionType": "addition" },
    { "levelId": 2, "difficulty": "medium", "questionType": "subtraction" },
    { "levelId": 3, "difficulty": "hard", "questionType": "multiplication" }
  ]
}
```

3. System Architecture Diagram

```
SYSTEM ARCHITECTURE DIAGRAM

                 JSON GAME CONFIG
                        ↓
                 CONFIG LOADER
                        ↓
                   GAME ENGINE
                  (CORE LOOP)
                        ↓
   INPUT HANDLER   GAME LOGIC      LEVEL MANAGER
                        ↓
                  SCORE MANAGER
                        ↓
                   RENDERER
                        ↓
                    WEB UI
```

4. Reusability Model

REUSABILITY MODEL
The engine separates game logic from game content using JSON-driven configuration.
REUSABLE GAME ENGINE

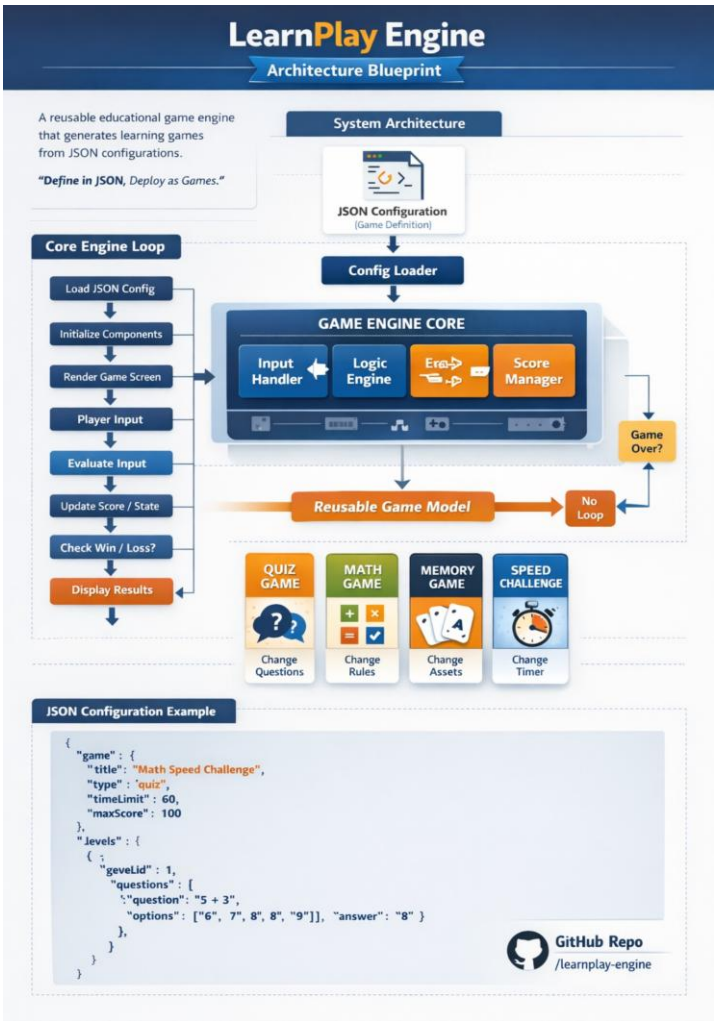
```
Math Game        Word Game        Memory Game
JSON Config      JSON Config      JSON Config
```

By replacing only the JSON configuration, new learning games can be generated without modifying the core engine.

5. GitHub Repository

GITHUB REPOSITORY

Repository Structure:

```
taptap-game-engine/
    │
    ├──── README.md
    │
    ├──── game-config.json
    │
    └──── engine-architecture.pdf
```

Repository Link:
https://github.com/kushbuchoudhary/taptap-game-Engine.git